

Yaxuan(Alice) Wen

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Research Interest

Systems Security, Operating Systems, Software Supply-Chain Security, Trustworthy Compilation

Education

New York University

Ph.D. in Computer Science

Brooklyn, NY

Aug 2024 – Present

- Advisor: Prof. Justin Cappos
- Research Interest: Compilation Security, Supply Chain Security, Operating Systems

New York University

M.Sc. in Cybersecurity

Brooklyn, NY

Aug 2022 – May 2024

- Advisor: Prof. Justin Cappos
- Research Interest: Supply Chain Security, Reproducible Build, System Security

Pennsylvania State University

B.Sc. in Computer Science

State College, PA

Aug 2017 – May 2021

Open-Source Leadership

Lind Project

2023–Present

Maintainer

- Maintain an open-source systems platform for running legacy POSIX applications in isolated compartments.
- Develop a userspace architecture for composing independently written system-call policies without kernel modifications or elevated privilege.
- Design mechanisms that allow isolated policy components to inspect, filter, transform, forward, or service system calls.
- Contribute to runtime infrastructure, system-call mediation, application compatibility, testing, and performance evaluation.

Projects

TriSeal: Attestable Closed-World Compilation

Lead Researcher

- Designed and implemented a prototype that treats compilation as a declared transformation over declared inputs, executed under a declared policy inside a remotely attestable hardware boundary.
- Built an in-boundary POSIX mediation layer with isolated storage and controlled ingress and egress, preventing undeclared host state from influencing compilation.
- Bound build outputs to the selected compiler, input manifest, execution policy, and measured environment through remote attestation and provenance metadata.
- Adapted a legacy C compiler toolchain to demonstrate compilation without relying on the host operating system, filesystem, or network for build semantics.

Lind-Project: Composable Userspace System-Call Policies

Lead Researcher

- Designed and implemented a userspace system-call mediation architecture that routes calls through independently developed, isolated policy components.
- Enabled policy components to inspect, filter, transform, forward, replicate, or directly service system calls while remaining isolated from applications and one another.
- Developed composition mechanisms for conditional routing and transparent replication, allowing policies to be assembled differently for individual workloads.

- Evaluated the architecture using filesystem policies, tracing, filtering, network mediation, compiler toolchains, and unmodified POSIX applications.
- Implemented a Linux-compatible system-call backend in Rust and evaluated its correctness and performance using microbenchmarks and application workloads.

Academic & Professional Experience

Secure Systems Lab, New York University

Brooklyn, New York

Graduate Research Assistant

May 2023 – Present

- Conduct research in systems security under Prof. Justin Cappos, with a focus on composable operating-system policies, trustworthy compilation, and WebAssembly-based isolation.
- Develop and maintain open-source research systems spanning runtime isolation, system-call mediation, software builds, and provenance.

Institute of Software, Chinese Academy of Science

Beijing, China

Research Assistant

June 2021 – Feb 2022

- Implemented a network simulation environment to model Software-Defined Networking (SDN) behavior under multi-user conditions.

Teaching Experience

Feb 2024 **Guest Lecture**, Advanced CS Topic: Supply Chain Security

NYU

Invited Talks

Nov 2025 - You Can Sign It, But Can You Trust It? Securing the Compilation Process

SecurityCon

Feb 2025 - Caging: An Abstraction For Finer-grained Isolation

NESysDay